ДЗ ML

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$$L(\theta) = 0{,}1\|\theta\|^2 + \frac{1}{N}\sum_{i=1}^{n} \max(0,\ 1 - y_i f(x_i;\theta))$$ ← нужно мин.

$$f(x,\theta) = x_1\theta_1 + x_2\theta_2 + \theta_3 \qquad \theta_{start} = (1\ 1\ 0)$$

i	0	1	2	3	4
X1	0	1	1	-0.5	0
X2	1	1	0	0.5	-0.5
Y	1	1	1	-1	-1

$$L(\theta) = 0{,}1(\theta_1^2 + \theta_2^2 + \theta_3^2) + \frac{1}{N}\sum_{i=1}^{N} \max(0,\ 1 - y_i(x_{1i}\theta_1 + x_{2i}\theta_2 + \theta_3))$$

$$\begin{cases} \dfrac{\partial L(\theta)}{\partial \theta_1} = 0{,}1 \cdot 2\theta_1 + \dfrac{1}{N}\sum_{i=1}^{N} M'(\theta_1) \\ \dfrac{\partial L(\theta)}{\partial \theta_2} = 0{,}1 \cdot 2\theta_2 + \dfrac{1}{N}\sum_{i=1}^{N} M'(\theta_2) \\ \dfrac{\partial L(\theta)}{\partial (\theta_3)} = 0{,}1 \cdot 2\theta_3 + \dfrac{1}{N} M_p'(\theta) \end{cases}$$

$$M(\theta) = \max(0,\ 1 - y_i(x_{1i}\theta_i + x_{2i}\theta_2 + \theta_3))$$

$$M'(\theta_n) = \begin{cases} 0, & \text{если } 1 - y_i(x_{11}\theta_i + x_{12}\theta_2 + x_{13}\theta_3) < 0 \\ -y_i x_{ni}, & \text{если } \text{"}-\text{"} \geq 0 \end{cases}$$

$$M_p'(\theta) = \begin{cases} 0, & \text{если } 1 - y_i(x_{1i}\theta_i + x_{2i}\theta_2 + \theta_3) < 0 \\ -y_i, & \text{если } \text{"}-\text{"} \geq 0 \end{cases}$$